

Measurement, Units, Dimensions and Vectors

The language of physics: how we measure, how we check our equations, and how we describe direction.

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Physics tries to describe nature with numbers and equations. Before we can write a single law of motion, we need three tools. The first is a reliable system of **units**, so that “5 metres” means the same thing everywhere. The second is **dimensional analysis**, a quick way to catch wrong equations and even to guess correct ones. The third is **vectors**, so we can describe quantities that point somewhere. This chapter builds all three, and finishes with the small amount of calculus the rest of the course relies on.

YOU WILL BE ABLE TO

- Use SI base units and prefixes, and convert between unit systems.
- Find the dimensions of any physical quantity and use the principle of homogeneity to check equations.
- Derive the form of a physical relation using dimensional analysis, and state the method's limitations.
- Make order-of-magnitude estimates.
- Apply significant-figure rules and propagate measurement errors.
- Resolve vectors into components and add vectors both graphically and by components.
- Compute dot and cross products and interpret them geometrically.
- Use derivatives and integrals to move between position, velocity and acceleration.

THINK FIRST

A friend tells you that the time t it takes a stone to fall a height h is $t = \sqrt{2hg}$, where g is the acceleration due to gravity. Without doing any physics, can you tell whether the formula is right?

Check the units. h is in metres and g in m/s^2 , so hg is in m^2/s^2 , and $\sqrt{hg}$ is in m/s . That is a speed, not a time, so the formula **must** be wrong. The correct result is $t = \sqrt{2h/g}$, which has units $\sqrt{\text{m}/(\text{m/s}^2)} = \text{s}$. Checking units like this is called *dimensional analysis*. It is the first tool in this chapter.

§1.1

Physics, models and measurement

Physics looks for a small number of general laws that explain a huge range of phenomena. The same law of gravitation describes a falling apple and the orbit of the Moon. The same laws of electromagnetism describe a fridge magnet and a radio wave.

To do this, physicists build **models**: simplified pictures that keep the features that matter and throw away the rest. When we treat a car as a “particle”, we ignore its size, shape and spinning wheels, because for the question “how long does it take to reach the next town?” none of those matter. A good model is as simple as possible while still giving accurate predictions. A large part of learning physics is learning *which* simplifications are safe.

Every model is tested against **measurement**. A measurement is a comparison with an agreed standard, the **unit**. So every measured quantity has two parts:

$$\text{physical quantity} = (\text{number}) \times (\text{unit}).$$

“The rod is 3” means nothing. “The rod is 3 m long” is a measurement. In this course we keep units attached to numbers at every step. They catch mistakes for free.

§1.2

The SI system of units

The *Système International d’Unités* (SI) is built on seven **base quantities**. All other quantities are **derived quantities**, defined as products and quotients of the base ones.

BASE QUANTITY	SI UNIT	SYMBOL	DIMENSION SYMBOL
Length	metre	m	L
Mass	kilogram	kg	M
Time	second	s	T
Electric current	ampere	A	I
Thermodynamic temperature	kelvin	K	Θ
Amount of substance	mole	mol	N
Luminous intensity	candela	cd	J

Since 2019, every SI unit has been defined by fixing the exact numerical value of a constant of nature. For example:

- The **second** is fixed by the frequency of a transition in the caesium-133 atom: exactly **9 192 631 770** oscillations.
- The **metre** is fixed by the speed of light, $c = 299\,792\,458\text{ m/s}$ *exactly*. So one metre is the distance light travels in $1/299\,792\,458$ of a second.
- The **kilogram** is fixed by Planck's constant, $h = 6.626\,070\,15 \times 10^{-34}\text{ J s}$ *exactly*.

This means any well-equipped laboratory anywhere in the universe could rebuild our units from scratch.

Prefixes

Prefixes scale a unit by powers of ten:

PREFIX	SYMBOL	FACTOR	PREFIX	SYMBOL	FACTOR
pico	p	10^{-12}	kilo	k	10^3
nano	n	10^{-9}	mega	M	10^6
micro	μ	10^{-6}	giga	G	10^9

PREFIX	SYMBOL	FACTOR	PREFIX	SYMBOL	FACTOR
milli	m	10^{-3}	tera	T	10^{12}
centi	c	10^{-2}	peta	P	10^{15}

Converting units

Treat units as algebraic quantities that multiply and cancel. Multiply by **conversion factors**: fractions equal to 1, such as $\frac{1000 \text{ m}}{1 \text{ km}}$.

WORKED EXAMPLE 1.1

A chain of conversions

(a) A car travels at 72 km/h. Express this speed in m/s. (b) The density of aluminium is 2.70 g/cm^3 . Express it in kg/m^3 .

Solution

(a) Multiply by conversion factors chosen so the unwanted units cancel:

$$72 \frac{\text{km}}{\text{h}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = \frac{72\,000}{3600} \text{ m/s} = 20 \text{ m/s}.$$

Useful shortcut: $1 \text{ km/h} = \frac{1000}{3600} \text{ m/s} = \frac{5}{18} \text{ m/s}$, and $72 \times \frac{5}{18} = 20$. ✓

(b) Both the mass unit and the length unit change. Remember that the length conversion must be **cubed**:

$$2.70 \frac{\text{g}}{\text{cm}^3} \times \frac{1 \text{ kg}}{1000 \text{ g}} \times \left(\frac{100 \text{ cm}}{1 \text{ m}} \right)^3 = 2.70 \times \frac{10^6}{10^3} \text{ kg/m}^3 = 2.70 \times 10^3 \text{ kg/m}^3.$$

Check. Water has density $1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3$, and aluminium is a little under three times as dense as water. ✓

COMMON MISTAKE

When converting areas or volumes, square or cube the *whole* conversion factor:
 $1 \text{ m}^2 = (100 \text{ cm})^2 = 10^4 \text{ cm}^2$, not 100 cm^2 .

IN PRACTICE

Unit errors are expensive. In 1999 NASA lost the \$125-million *Mars Climate Orbiter*. One team’s software reported thruster impulse in pound-force seconds, and the navigation software expected newton seconds, a factor of 4.45. In 1983 an Air Canada Boeing 767, the “Gimli Glider”, ran out of fuel in flight because fuel had been loaded in pounds instead of kilograms. In hospitals, confusing mg with μg (a factor of 1000) or mL/h with mL/min (a factor of 60) is a recognised cause of serious dosing errors. Writing units at every step of every calculation is professional practice, not pedantry.

§1.3

Dimensions and dimensional analysis

The **dimensions** of a quantity tell us how it is built from the base quantities, regardless of the units used. We write $[Q]$ for “the dimensions of Q ”. For example, speed is a length divided by a time, whether it is measured in m/s, km/h or furlongs per fortnight, so

$$[v] = \frac{[L]}{[T]} = [LT^{-1}].$$

To find the dimensions of any quantity, start from an equation that defines it and work back to base quantities.

QUANTITY	DEFINING RELATION	DIMENSIONS	SI UNIT
Area, volume	ℓ^2, ℓ^3	$[L^2], [L^3]$	m^2, m^3
Velocity	$\Delta x / \Delta t$	$[LT^{-1}]$	m/s
Acceleration	$\Delta v / \Delta t$	$[LT^{-2}]$	m/s^2
Force	ma	$[MLT^{-2}]$	newton, N
Work, energy	Fd	$[ML^2T^{-2}]$	joule, J
Power	W/t	$[ML^2T^{-3}]$	watt, W

QUANTITY	DEFINING RELATION	DIMENSIONS	SI UNIT
Momentum, impulse	$mv, F\Delta t$	$[MLT^{-1}]$	kg m/s
Pressure, stress	F/A	$[ML^{-1}T^{-2}]$	pascal, Pa
Frequency	$1/T$	$[T^{-1}]$	hertz, Hz
Angle	arc/radius	dimensionless	rad
Gravitational constant G	$F r^2 / (m_1 m_2)$	$[M^{-1}L^3T^{-2}]$	N m ² /kg ²
Planck's constant h	E/ν	$[ML^2T^{-1}]$	J s

KEY IDEA

Principle of homogeneity. In a physically correct equation, every term has the same dimensions. You can only add, subtract or equate quantities with identical dimensions. The arguments of **sin**, **cos**, **exp** and **ln** must be dimensionless. So in $x = A \sin(\omega t)$, the product ωt is dimensionless, which forces $[\omega] = [T^{-1}]$.

Dimensional analysis has three main uses.

Use 1: checking equations

If the terms of an equation do not all have the same dimensions, the equation is wrong. If they do, it *might* be right. Dimensional analysis cannot detect a wrong numerical factor such as $\frac{1}{2}$ or 2π .

WORKED EXAMPLE 1.2**Is this equation possible?**

A student writes the distance travelled under constant acceleration as $s = ut + \frac{1}{2}at^3$, where u is the initial speed. Check the equation dimensionally.

Solution

Check each term separately:

- $[s] = [L]$

- $[ut] = [LT^{-1}][T] = [L] \checkmark$
- $[\frac{1}{2}at^3] = [LT^{-2}][T^3] = [LT] \times$

The last term has dimensions $[LT]$, not $[L]$, so the equation is **wrong**. Changing t^3 to t^2 fixes the dimensions: $[at^2] = [L]$. The factor $\frac{1}{2}$ is a pure number, so dimensional analysis can neither confirm nor rule it out.

Use 2: converting units between systems

If a quantity has dimensions $[M^a L^b T^c]$, then its numerical value changes between unit systems as

$$n_2 = n_1 \left(\frac{M_1}{M_2} \right)^a \left(\frac{L_1}{L_2} \right)^b \left(\frac{T_1}{T_2} \right)^c,$$

where M_1, L_1, T_1 are the old units and M_2, L_2, T_2 the new ones. This works because the *physical* quantity stays the same: $n_1 u_1 = n_2 u_2$.

WORKED EXAMPLE 1.3

The gravitational constant in CGS units

In SI units, $G = 6.67 \times 10^{-11} \text{ N m}^2/\text{kg}^2$. Find its value in CGS units (gram, centimetre, second).

Solution

From the table, $[G] = [M^{-1} L^3 T^{-2}]$, so $a = -1$, $b = 3$, $c = -2$. The unit ratios are

$$\frac{M_1}{M_2} = \frac{1 \text{ kg}}{1 \text{ g}} = 10^3, \quad \frac{L_1}{L_2} = \frac{1 \text{ m}}{1 \text{ cm}} = 10^2, \quad \frac{T_1}{T_2} = 1.$$

Then

$$n_2 = 6.67 \times 10^{-11} (10^3)^{-1} (10^2)^3 (1)^{-2} = 6.67 \times 10^{-11} \times 10^{-3} \times 10^6 = 6.67 \times 10^{-8}.$$

So $G = 6.67 \times 10^{-8} \text{ cm}^3 \text{ g}^{-1} \text{ s}^{-2}$ (equivalently, $\text{dyn cm}^2/\text{g}^2$).

Use 3: deriving relations

If we can guess which quantities a result depends on, dimensional analysis often fixes the form of the relation, up to a dimensionless constant.

WORKED EXAMPLE 1.4

The period of a pendulum

The period T of a simple pendulum might depend on the mass m of the bob, the length ℓ of the string and the gravitational acceleration g . Find the form of the relation.

Solution

Set up. Assume $T = k m^a \ell^b g^c$, where k is a dimensionless constant.

Execute. Equate dimensions on both sides:

$$[M^0 L^0 T^1] = [M]^a [L]^b [LT^{-2}]^c = [M^a L^{b+c} T^{-2c}].$$

Matching the powers of each base dimension:

$$\begin{aligned} M : \quad a &= 0, \\ T : \quad -2c &= 1 \Rightarrow c = -\frac{1}{2}, \\ L : \quad b + c &= 0 \Rightarrow b = \frac{1}{2}. \end{aligned}$$

Therefore

$$T = k \sqrt{\frac{\ell}{g}}.$$

Evaluate. The period does **not** depend on the mass. This is a genuine physical prediction, and it is true. A full analysis (Chapter 9) gives $k = 2\pi$ for small swings.

NOTE

Limitations of dimensional analysis.

1. It cannot find dimensionless constants such as 2π or $\frac{1}{2}$.
2. It fails if the result depends on a dimensionless quantity, such as an angle. The exact pendulum period depends on the amplitude θ_0 , and dimensional analysis cannot tell us how.
3. It cannot handle sums of terms such as $s = ut + \frac{1}{2}at^2$ in a single step.
4. With three base dimensions we get only three equations, so we can solve for at most three unknown exponents.
5. It cannot distinguish quantities that share dimensions, such as work and torque.

IN PRACTICE

Dimensionless numbers run engineering. Engineers group variables into dimensionless combinations and design with those. The most famous is the **Reynolds number**

$$Re = \frac{\rho v D}{\eta},$$

where ρ is the fluid's density, v its speed, D the pipe diameter and η the viscosity. It decides whether flow is smooth (laminar, $Re \lesssim 2000$) or turbulent ($Re \gtrsim 4000$), in an oil pipeline and in an artery alike (Problem P1.17). Because only dimensionless groups matter, a 1:20 scale model of a car or bridge in a wind tunnel behaves like the full-size structure, *provided* the dimensionless numbers match. In physiology, scaling arguments explain why a mouse's heart beats about 600 times a minute and an elephant's about 30. They also underlie dosing drugs by body-surface area rather than by mass.

§1.4

Order-of-magnitude estimates

Physicists often need a rough answer quickly: is the effect 10^{-3} or 10^3 ? An **order-of-magnitude estimate** keeps only the nearest power of ten. Make reasonable round-number assumptions, combine them, and keep one significant figure. These are often called *Fermi problems*, after Enrico Fermi, who was famous for them.

WORKED EXAMPLE 1.5

How many heartbeats in a lifetime?

Estimate the number of times a human heart beats in a lifetime.

Solution

Assumptions: about 70 beats per minute and a lifetime of about 80 years.

$$\text{minutes in a lifetime} \approx 80 \text{ yr} \times 365 \frac{\text{d}}{\text{yr}} \times 24 \frac{\text{h}}{\text{d}} \times 60 \frac{\text{min}}{\text{h}} \approx 4.2 \times 10^7 \text{ min.}$$

$$\text{beats} \approx 70 \times 4.2 \times 10^7 \approx 3 \times 10^9.$$

A lifetime is about **three billion heartbeats**, so the order of magnitude is 10^9 . Changing the assumptions slightly (60 or 80 beats per minute, 70 or 90 years) does not change the power of ten.

§1.5

Significant figures and errors

Significant figures

No measurement is exact. The **significant figures** of a number are all the reliably known digits plus the first uncertain one. A ruler with millimetre markings might give **12.3** cm. Here the 3 is the uncertain digit, and writing **12.300** cm would claim a precision we do not have.

Rules for counting significant figures:

- All non-zero digits are significant: **4.37** has 3.
- Zeros *between* non-zero digits are significant: **2.05** has 3.
- Leading zeros are not significant. They only locate the decimal point: **0.0042** has 2.
- Trailing zeros after a decimal point are significant: **3.50** has 3.
- Trailing zeros in a whole number are ambiguous. Use scientific notation: **1.5×10^3** (2 s.f.) or **1.500×10^3** (4 s.f.).

Rules for calculations:

- **Multiplying or dividing:** keep as many significant figures as the *least* precise factor. For example, **$3.2 \times 1.234 = 3.9$** (2 s.f.).
- **Adding or subtracting:** keep as many *decimal places* as the term with the fewest. For example, **$12.1 + 0.035 = 12.1$** .

Absolute and relative error

If a quantity is measured as $A \pm \Delta A$, then ΔA is the **absolute error**, $\Delta A/A$ the **relative (fractional) error**, and $100 \Delta A/A$ the **percentage error**.

The maximum error in a calculated result follows from two rules:

- **Sum or difference:** for $Z = A \pm B$, absolute errors add: **$\Delta Z = \Delta A + \Delta B$** .
- **Product, quotient or power:** for $Z = A^p B^q / C^r$, *relative* errors add, each weighted by the size of its power:

$$\frac{\Delta Z}{Z} = |p| \frac{\Delta A}{A} + |q| \frac{\Delta B}{B} + |r| \frac{\Delta C}{C}.$$

These rules give the worst case. When errors are random and independent, a better estimate combines them in quadrature, $\Delta Z/Z = \sqrt{(p \Delta A/A)^2 + \dots}$. You will meet this in the laboratory course.

WORKED EXAMPLE 1.6

Measuring g with a pendulum

A student measures the length of a pendulum as $\ell = 1.00 \pm 0.01$ m and its period as $T = 2.00 \pm 0.02$ s. Using $g = 4\pi^2\ell/T^2$, find g and its uncertainty. Which measurement should the student improve?

Solution

Central value:

$$g = \frac{4\pi^2(1.00 \text{ m})}{(2.00 \text{ s})^2} = \frac{39.48}{4.00} \text{ m/s}^2 = 9.87 \text{ m/s}^2.$$

Relative error. Here $g \propto \ell^1 T^{-2}$:

$$\frac{\Delta g}{g} = 1 \cdot \frac{\Delta \ell}{\ell} + 2 \cdot \frac{\Delta T}{T} = \frac{0.01}{1.00} + 2 \times \frac{0.02}{2.00} = 0.01 + 0.02 = 0.03 \text{ (3\%)}.$$

Absolute error: $\Delta g = 0.03 \times 9.87 \approx 0.3 \text{ m/s}^2$. So

$$g = (9.9 \pm 0.3) \text{ m/s}^2.$$

Improvement. The period contributes 2% of the 3%, because it is squared. The student should time many oscillations, say 20, and divide by 20. The reaction-time error is then spread over the whole run, reducing $\Delta T/T$ by a factor of about 20.

§1.6

Scalars and vectors

A **scalar** is a quantity fully described by a single number with a unit: mass, time, temperature, energy, charge. A **vector** has both **magnitude** and **direction**, *and* combines with other vectors by the triangle law of addition: displacement, velocity, acceleration, force, momentum.

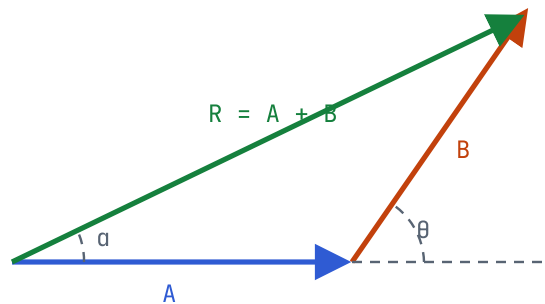
COMMON MISTAKE

Having a direction is not enough to make something a vector. Electric current has a direction along a wire, yet two currents of 3 A and 4 A flowing into a junction give 7 A whatever the angle between the wires. Current adds like a scalar, so it is a scalar. A quantity is a vector only if it adds like a displacement.

We write vectors in bold italic with an arrow, $\vec{A}$, and the magnitude as A or $|\vec{A}|$. A magnitude is never negative.

Adding vectors geometrically

To add $\vec{A}$ and $\vec{B}$, place the tail of $\vec{B}$ at the head of $\vec{A}$. The resultant $\vec{R} = \vec{A} + \vec{B}$ runs from the tail of $\vec{A}$ to the head of $\vec{B}$. This is the **triangle law**. Equivalently, draw both vectors from the same point and complete the parallelogram. The diagonal is the resultant (the **parallelogram law**).



Triangle law: put B's tail at A's head. The resultant R runs from A's tail to B's head. θ is the angle between A and B.

Applying the law of cosines to this triangle, where the interior angle at the head of $\vec{A}$ is $180^\circ - \theta$, gives the magnitude of the resultant:

$$R = \sqrt{A^2 + B^2 + 2AB \cos \theta}.$$

Its angle α from $\vec{A}$ satisfies

$$\tan \alpha = \frac{B \sin \theta}{A + B \cos \theta}.$$

Special cases worth remembering:

- $\theta = 0$ (parallel): $R = A + B$, the maximum.
- $\theta = 180^\circ$ (antiparallel): $R = |A - B|$, the minimum.
- $\theta = 90^\circ$: $R = \sqrt{A^2 + B^2}$.

In general $|A - B| \leq R \leq A + B$.

Subtraction is addition of the negative: $\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$, where $-\vec{B}$ has the same magnitude as $\vec{B}$ and points the opposite way. Its magnitude is $|\vec{A} - \vec{B}| = \sqrt{A^2 + B^2 - 2AB \cos \theta}$.

Components and unit vectors

Geometric addition is clumsy for more than two vectors. Components are far more powerful.

A **unit vector** has magnitude 1 and only indicates direction. The Cartesian unit vectors $\hat{i}$, $\hat{j}$, $\hat{k}$ point along $+x$, $+y$, $+z$. Any vector can be written as

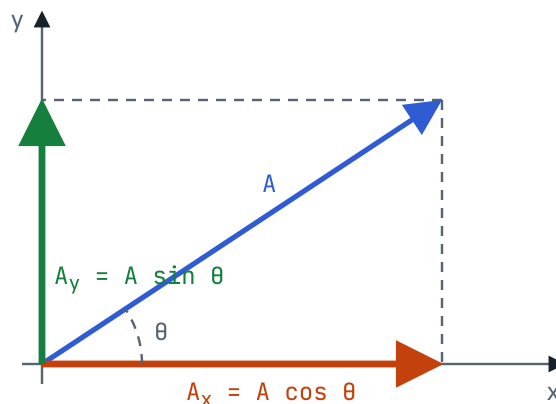
$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k},$$

where the **components** A_x , A_y , A_z are scalars and can be negative. In two dimensions, a vector of magnitude A at angle θ measured counter-clockwise from the $+x$ axis has

$$A_x = A \cos \theta, \quad A_y = A \sin \theta,$$

and conversely

$$A = \sqrt{A_x^2 + A_y^2}, \quad \tan \theta = \frac{A_y}{A_x}.$$



Resolving a vector: the x component is $A \cos \theta$ and the y component is $A \sin \theta$, with θ measured from the $+x$ axis.

COMMON MISTAKE

The formula $\tan \theta = A_y/A_x$ alone cannot tell apart angles that differ by 180° . For example, $(-3, -4)$ and $(3, 4)$ give the same ratio. Always check the signs of the components to find the correct quadrant.

Adding by components. To add vectors, add their components:

$$\vec{R} = \vec{A} + \vec{B} + \dots \implies R_x = A_x + B_x + \dots, \quad R_y = A_y + B_y + \dots$$

WORKED EXAMPLE 1.7**Adding two forces**

Two forces of 3.0 N and 4.0 N act on a particle, with 60° between them. Find the magnitude of the resultant and its direction relative to the 3.0 N force.

Solution

Use the resultant formula with $A = 3.0$ N, $B = 4.0$ N and $\theta = 60^\circ$:

$$R = \sqrt{3.0^2 + 4.0^2 + 2(3.0)(4.0) \cos 60^\circ} = \sqrt{9 + 16 + 12} = \sqrt{37} = 6.1 \text{ N.}$$

Direction relative to the 3.0 N force:

$$\tan \alpha = \frac{4.0 \sin 60^\circ}{3.0 + 4.0 \cos 60^\circ} = \frac{3.464}{5.0} = 0.693 \implies \alpha = 34.7^\circ.$$

Check. R lies between $|A - B| = 1$ N and $A + B = 7$ N. ✓ The resultant leans towards the larger force: 34.7° from the 3 N force, so 25.3° from the 4 N force. ✓

WORKED EXAMPLE 1.8**Three displacements by components**

A hiker makes three straight-line walks: $A = 10.0$ km at 30° , then $B = 15.0$ km at 135° , then $C = 8.0$ km at 270° . All angles are measured counter-clockwise from east. Find the hiker's net displacement.

Solution

Set up. Take $+x$ as east and $+y$ as north, and tabulate the components.

VECTOR	x COMPONENT (KM)	y COMPONENT (KM)
$\vec{A}$	$10.0 \cos 30^\circ = 8.660$	$10.0 \sin 30^\circ = 5.000$
$\vec{B}$	$15.0 \cos 135^\circ = -10.607$	$15.0 \sin 135^\circ = 10.607$
$\vec{C}$	$8.0 \cos 270^\circ = 0$	$8.0 \sin 270^\circ = -8.000$
$\vec{R}$	-1.947	7.607

Execute.

$$R = \sqrt{(-1.947)^2 + (7.607)^2} = \sqrt{3.79 + 57.87} = 7.85 \text{ km.}$$

$\tan^{-1}(7.607/1.947) = 75.6^\circ$. Since $R_x < 0$ and $R_y > 0$, the vector lies in the second quadrant, so

$$\theta = 180^\circ - 75.6^\circ = 104.4^\circ \text{ from east,}$$

which is 14.4° west of north.

Evaluate. A calculator returns $\tan^{-1}(7.607/(-1.947)) = -75.6^\circ$, which is in the wrong quadrant. This is exactly the warning above.

IN PRACTICE

Resolving forces is how structures are designed. The members of a roof truss, the cables of a suspension bridge and the tendons around your knee all carry forces along fixed directions. Engineers resolve every load into components along and across each member, then require the components at each joint to sum to zero. In biomechanics, the same method shows why the patellar tendon can carry several times body weight when you climb stairs. Vectors also underlie navigation. A GPS receiver finds your position vector from its distances to four or more satellites, and an aircraft's velocity over the ground is the vector sum of its airspeed and the wind velocity.

§1.7

Multiplying vectors

There are two useful ways to multiply two vectors. One gives a scalar, the other a vector.

The scalar (dot) product

$$\vec{A} \cdot \vec{B} = AB \cos \theta = A_x B_x + A_y B_y + A_z B_z.$$

Geometrically, $\vec{A} \cdot \vec{B}$ is A times the projection of $\vec{B}$ onto the direction of $\vec{A}$, which is $B \cos \theta$.

Properties:

- It is commutative: $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$.
- It is distributive: $\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$.
- $\vec{A} \cdot \vec{A} = A^2$.
- $\vec{A} \cdot \vec{B} = 0$ for non-zero vectors means $\vec{A} \perp \vec{B}$.
- $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$ and $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$. The component formula follows from these.

The component of $\vec{B}$ along $\vec{A}$ is $\vec{B} \cdot \hat{A} = \frac{\vec{A} \cdot \vec{B}}{A}$. Physical example: the work done by a force, $W = \vec{F} \cdot \vec{d}$.

The vector (cross) product

$\vec{C} = \vec{A} \times \vec{B}$ is a vector with:

- magnitude $C = AB \sin \theta$, where $0 \leq \theta \leq 180^\circ$;
- direction perpendicular to the plane containing $\vec{A}$ and $\vec{B}$, given by the **right-hand rule**: point the fingers of your right hand along $\vec{A}$ and curl them towards $\vec{B}$ through the smaller angle. Your thumb points along $\vec{C}$.

In components, the cross product is the “determinant”:

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = (A_y B_z - A_z B_y) \hat{i} - (A_x B_z - A_z B_x) \hat{j} + (A_x B_y - A_y B_x) \hat{k}.$$

Properties:

- It is anti-commutative: $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$.
- It is distributive over addition.
- $\vec{A} \times \vec{A} = \vec{0}$, and parallel vectors have zero cross product.
- $\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$ (cyclic order). Reversing the order flips the sign.
- $|\vec{A} \times \vec{B}|$ equals the area of the parallelogram with sides $\vec{A}$ and $\vec{B}$.

Physical examples: torque $\vec{\tau} = \vec{r} \times \vec{F}$, angular momentum $\vec{L} = \vec{r} \times \vec{p}$, and the magnetic force $\vec{F} = q\vec{v} \times \vec{B}$.

WORKED EXAMPLE 1.9

Dot and cross products

Given $\vec{A} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{B} = \hat{i} - \hat{j} + 2\hat{k}$, find (a) the angle between them, (b) $\vec{A} \times \vec{B}$, (c) a unit vector perpendicular to both, and (d) the area of the parallelogram they span.

Solution

(a) The dot product and the magnitudes:

$$\begin{aligned}\vec{A} \cdot \vec{B} &= (2)(1) + (3)(-1) + (-1)(2) = -3, \\ A &= \sqrt{4 + 9 + 1} = \sqrt{14}, \quad B = \sqrt{1 + 1 + 4} = \sqrt{6}. \\ \cos \theta &= \frac{-3}{\sqrt{14}\sqrt{6}} = \frac{-3}{\sqrt{84}} = -0.327 \Rightarrow \theta = 109.1^\circ.\end{aligned}$$

A negative dot product always means an obtuse angle.

(b) Expand the determinant term by term:

$$\begin{aligned}\vec{A} \times \vec{B} &= [(3)(2) - (-1)(-1)]\hat{i} - [(2)(2) - (-1)(1)]\hat{j} + [(2)(-1) - (3)(1)]\hat{k} \\ &= 5\hat{i} - 5\hat{j} - 5\hat{k}.\end{aligned}$$

Check: the cross product must be perpendicular to both vectors.

$$(5, -5, -5) \cdot (2, 3, -1) = 10 - 15 + 5 = 0 \checkmark \text{ and } (5, -5, -5) \cdot (1, -1, 2) = 5 + 5 - 10 = 0 \checkmark.$$

(c) The magnitude is $|\vec{A} \times \vec{B}| = 5\sqrt{3}$, so

$$\hat{n} = \frac{\vec{A} \times \vec{B}}{|\vec{A} \times \vec{B}|} = \frac{1}{\sqrt{3}}(\hat{i} - \hat{j} - \hat{k}).$$

The opposite vector $-\hat{n}$ is also perpendicular to both.

(d) Area = $|\vec{A} \times \vec{B}| = 5\sqrt{3} \approx 8.66$ square units.

Cross-check using (a): $AB \sin \theta = \sqrt{84} \sin 109.1^\circ = 9.165 \times 0.945 = 8.66. \checkmark$

§1.8

Calculus tools for physics

Motion is change, and calculus is the mathematics of change. This course uses only a few ideas, collected here.

The derivative: instantaneous rate of change

The derivative $\frac{dy}{dx}$ is the slope of the graph of $y(x)$: the limit of $\Delta y / \Delta x$ as $\Delta x \rightarrow 0$. In mechanics, **velocity** is the rate of change of position and **acceleration** is the rate of change of velocity:

$$v = \frac{dx}{dt}, \quad a = \frac{dv}{dt} = \frac{d^2x}{dt^2}.$$

FUNCTION	DERIVATIVE
t^n	$n t^{n-1}$
$\sin \omega t$	$\omega \cos \omega t$
$\cos \omega t$	$-\omega \sin \omega t$
e^{kt}	$k e^{kt}$
$\ln t$	$1/t$

The chain rule, $\frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt}$, gives an extremely useful form of the acceleration:

$$a = \frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt} = v \frac{dv}{dx}.$$

Use this whenever acceleration is given as a function of **position**.

Maxima and minima

A smooth function $y(x)$ has a local maximum or minimum where $dy/dx = 0$. It is a maximum if $d^2y/dx^2 < 0$ and a minimum if $d^2y/dx^2 > 0$.

The integral: accumulated change

Integration reverses differentiation and adds up infinitesimal contributions. Geometrically, the definite integral is the area under a curve:

$$x(t) = x_0 + \int_0^t v(t') dt', \quad v(t) = v_0 + \int_0^t a(t') dt'.$$

Useful results: $\int t^n dt = \frac{t^{n+1}}{n+1} + C$ for $n \neq -1$, and $\int \frac{dt}{t} = \ln t + C$.

Separation of variables. If $\frac{dv}{dt} = f(v)$, rearrange to $\frac{dv}{f(v)} = dt$ and integrate both sides. We use this repeatedly, for example for motion with air resistance and for charging capacitors.

WORKED EXAMPLE 1.10

From position to velocity and acceleration

A particle moves along the x -axis with $x(t) = 2t^3 - 9t^2 + 12t$, where x is in metres and t in seconds. Find (a) $v(t)$ and $a(t)$, (b) when the particle is momentarily at rest and where it is then, and (c) the total distance travelled from $t = 0$ to $t = 3$ s.

Solution

(a) Differentiate once for velocity and again for acceleration:

$$v = \frac{dx}{dt} = 6t^2 - 18t + 12 \text{ m/s}, \quad a = \frac{dv}{dt} = 12t - 18 \text{ m/s}^2.$$

(b) The particle is at rest when $v = 0$:

$$6(t^2 - 3t + 2) = 6(t - 1)(t - 2) = 0 \Rightarrow t = 1 \text{ s or } t = 2 \text{ s}.$$

The positions are $x(1) = 2 - 9 + 12 = 5$ m and $x(2) = 16 - 36 + 24 = 4$ m. At $t = 1$ s, $a = -6 \text{ m/s}^2 < 0$, so x has a local maximum: the particle turns back. At $t = 2$ s, $a = +6 \text{ m/s}^2$, so it turns forward again.

(c) Distance is **not** $|x(3) - x(0)|$ when the particle reverses. Add up each leg separately. With $x(0) = 0$ and $x(3) = 54 - 81 + 36 = 9$ m:

- $0 \rightarrow 1$ s: from 0 to 5 m, a distance of 5 m.

- $1 \rightarrow 2$ s: from 5 to 4 m, a distance of 1 m.
- $2 \rightarrow 3$ s: from 4 to 9 m, a distance of 5 m.

Total distance = $5 + 1 + 5 = 11$ m. The displacement is only 9 m.

CHAPTER SUMMARY

- Every measurement is a number times a unit. SI has seven base units, each now defined through a fixed constant of nature.
- Dimensions express a quantity in terms of $M, L, T, \dots$ **Homogeneity**: every term in a valid equation has the same dimensions, and the arguments of **sin**, **exp** and **ln** are dimensionless.
- Dimensional analysis checks equations, converts units ($n_2 = n_1(M_1/M_2)^a(L_1/L_2)^b(T_1/T_2)^c$) and derives relations up to a dimensionless constant.
- Errors: absolute errors add for sums and differences. Relative errors, weighted by powers, add for products, quotients and powers.
- Vectors add by the triangle law: $R = \sqrt{A^2 + B^2 + 2AB \cos \theta}$. Adding by components is usually easier: $R_x = \sum A_x$, $R_y = \sum A_y$. Check the quadrant.
- $\vec{A} \cdot \vec{B} = AB \cos \theta = \sum A_i B_i$ is a scalar. $\vec{A} \times \vec{B}$ has magnitude $AB \sin \theta$, is perpendicular to both by the right-hand rule, and is anti-commutative.
- $v = dx/dt$ and $a = dv/dt = v dv/dx$. Integrals reverse derivatives and represent areas under curves.

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P1.1

Which of the following pairs have the same dimensions? (a) work and torque, (b) impulse and momentum, (c) pressure and energy density, (d) angular momentum and Planck's constant. Justify each answer.

P1.2

Can the sum of two vectors of unequal magnitude ever be zero? Can the sum of three vectors of unequal magnitude be zero? Explain.

P1.3

Round each number to three significant figures, and state how many significant figures it originally had: (a) **0.004 5670**, (b) **2.0063**, (c) **6.0225×10^{23}** .

P1.4

Two non-zero vectors have both $\vec{A} \cdot \vec{B} = 0$ and $\vec{A} \times \vec{B} = \vec{0}$. Is this possible? Explain.

LEVEL B – STANDARD PROBLEMS**P1.5**

The van der Waals equation for a real gas is $\left(P + \frac{a}{V^2}\right)(V - b) = RT$, where P is pressure and V is volume. Find the dimensions of a and b .

P1.6

The speed v of a transverse wave on a stretched string depends on the tension F in the string and its mass per unit length μ . Use dimensional analysis to find the form of v .

P1.7

A man walks 30 m north, then 20 m east, then $30\sqrt{2}$ m south-west. Find his final displacement from the starting point, giving both magnitude and direction.

P1.8

Find the scalar component of $\vec{A} = 3\hat{i} + 4\hat{j}$ along the direction of $\vec{B} = \hat{i} + \hat{j}$. Also find the vector component of $\vec{A}$ along $\vec{B}$.

P1.9

The resistance of a wire is found from $R = V/I$, with $V = (100 \pm 5)$ V and $I = (10 \pm 0.2)$ A. Find R , its percentage error and its absolute error.

P1.10

Two vectors have magnitudes 5 units and 12 units, and their sum has magnitude 13 units. Find the angle between them and the magnitude of their difference.

P1.11

Estimate the mass of air in a classroom that measures $10\text{ m} \times 8\text{ m} \times 3\text{ m}$. The density of air is about 1.2 kg/m^3 . Compare your answer with the mass of a typical student.

P1.12

A patient of mass 70 kg is prescribed dopamine at $5.0\text{ }\mu\text{g kg}^{-1}\text{ min}^{-1}$. The infusion bag contains 400 mg of dopamine in 250 mL of saline. At what rate, in mL/h, must the infusion pump be set? If a nurse mistakenly enters the rate in mL/min instead of mL/h, by what factor is the patient overdosed?

LEVEL C – CHALLENGE PROBLEMS**P1.13**

In a new system of units, the unit of mass is $\alpha\text{ kg}$, the unit of length is $\beta\text{ m}$ and the unit of time is $\gamma\text{ s}$. What is the numerical value of 1 joule in this system?

P1.14

Find the area of the triangle with vertices $P(1, 0, 0)$, $Q(0, 2, 0)$ and $R(0, 0, 3)$, with coordinates in metres. Also find the unit vector normal to the plane of the triangle.

P1.15

A particle moves along the x -axis with velocity $v = \beta\sqrt{x}$, where β is a positive constant. It starts from $x = 0$ at $t = 0$. Find (a) $x(t)$, (b) $v(t)$ and $a(t)$, and (c) the average velocity over the first s metres of motion.

P1.16

The frequency f of oscillation of a small liquid drop is thought to depend on its radius r , the density ρ of the liquid and its surface tension S (force per unit length). Use dimensional analysis to find how f depends on these quantities. By what factor does the frequency change if the radius is doubled?

P1.17

(a) Show that the Reynolds number $Re = \rho v D / \eta$ is dimensionless, given that viscosity η has SI unit Pa s. (b) Blood ($\rho = 1060 \text{ kg/m}^3$, $\eta = 3.5 \times 10^{-3} \text{ Pa s}$) flows through an aorta of diameter 2.5 cm. At rest the mean speed is 0.30 m/s. During heavy exercise it rises to 1.2 m/s. Find Re in each case and comment on whether the flow is likely to be laminar or turbulent. (c) A water pipe in a building ($\eta = 1.0 \times 10^{-3} \text{ Pa s}$) has diameter 2.0 cm. Above what speed would you expect turbulence, taking $Re \approx 2000$ as the threshold?

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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